



SnakeDash: Dash Through The Challenge, Eating Grades To Stay Ahead

**Presented to the
College of Engineering and Architecture
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Gen. Douglas MacArthur Highway, Davao City**

**In Partial Fulfillment
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CPE144L Mobile Application Development (Laboratory)**

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Brief Description of the Mobile Application

SnakeDash reinterprets the classic "Snake" game by introducing a unique academic theme. Players control student characters and pursue grades, symbolized as "fruits." The game combines nostalgic gameplay with humor and educational elements, offering a compelling experience for students. It aims to fill the gap in the mobile gaming market for theme-based educational games, providing both entertainment and a connection to academic life.

Requirements Checklist

The application provides an engaging experience with functionalities such as user authentication, dynamic game progression, and real-time leaderboard integration. Specifically, this includes:

FUNCTIONAL REQUIREMENTS	IMPLEMENTATION
1. User Registration and Login Authentication	Firestore Authentication is utilized to handle user registration, login, session management. Secure login credentials are verified and stored, ensuring a smooth authentication process.
2. Scoring System with Leaderboard Integration	A scoring system is implemented to track player performance, with scores stored in a Firestore database. The leaderboard is dynamically updated, displaying the top players based on their scores.
3. Snake Customization	The game allows users to customize the appearance of their snake by selecting different sprites for the snake's head. This feature is integrated into the user interface using Jetpack Compose.
4. Academic-themed Items	Academic-themed items are introduced as collectible objects in the game, replacing traditional fruits. These items are dynamically displayed and consumed by the snake during gameplay.
5. Game Progression	As the game advances, the snake's speed increases, progressively enhancing the difficulty. This functionality is controlled through game

	logic and is designed to provide a challenging experience for the player.
6. Log Out Functionality	A secure log-out feature is integrated using Firebase, allowing users to sign out from the application, ensuring that their session ends safely.
7. Screen Redirection	Screen navigation is handled through button clicks, using explicit intents to redirect users to different screens, providing a smooth transition between game modes.
8. Multiplayer Mode	The multiplayer mode is implemented to enable real-time gameplay between users. Firebase's real-time database supports synchronization and communication between players, allowing them to compete in a dynamic, multi-user environment.

Table 1: Functional Requirements

NON-FUNCTIONAL REQUIREMENTS	IMPLEMENTATION
1. Overall Responsiveness of the Game's Screen	The game is designed to be responsive, adjusting its layout and controls to different screen sizes, ensuring a smooth experience across various devices.
2. Secure User Data and Credentials	User data and credentials are securely managed using Firebase Authentication, ensuring that login information is encrypted and stored safely.
3. Usability and Accessibility	The game's interface is user-friendly and easy to navigate, with accessible design features like high contrast for visibility and compatibility with screen readers for users with disabilities.

Table 2: Non-functional Requirements

TECHNICAL REQUIREMENTS	IMPLEMENTATION
1. Platform	The platform used for developing the game is Android Studio, which provides a comprehensive environment for building Android applications. Android Studio offers the necessary tools and features to facilitate efficient app development, ensuring compatibility with a wide range of Android devices.
2. Programming Language	Kotlin is chosen for its conciseness, compatibility with Android, and its ability to streamline development by reducing boilerplate code, making it ideal for building robust and scalable applications.
3. Libraries	Firestore is used for user authentication and managing data, providing a secure and reliable backend solution. Jetpack Compose is utilized to build the user interface, enabling a modern, declarative approach to creating responsive and visually appealing layouts.

Table 3: Technical Requirements

Use Cases

The use cases outline the key screens and functionalities within the SnakeDash game, focusing on user login, gameplay, and multiplayer interactions. Each case describes the actions and system responses, ensuring a smooth and engaging user experience from start to finish. Specifically, this includes:

CASE	DESCRIPTION
1. Landing Screen	This contains the first screen that the user will see ("click anywhere to start"). It has a background image and the game's logo.
2. Login Screen	After the Landing Screen, users will enter their username and password. The system

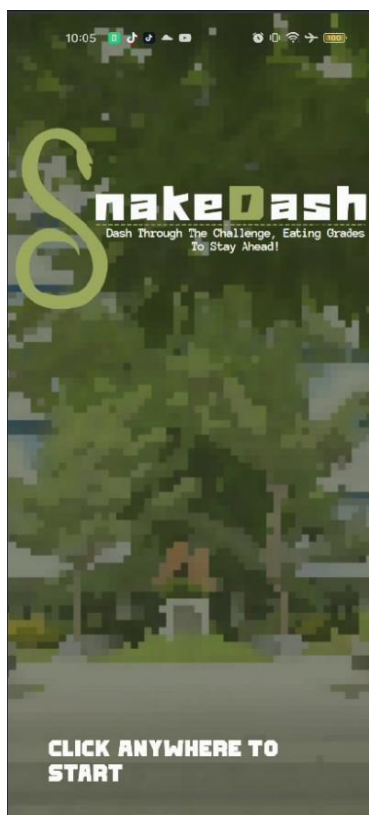
	checks if their credentials match what's stored in the database.
3. Sign Up Screen	If the user hasn't registered yet, they will enter their name, username, and password. Once registered, the user will go back to the login screen,, and from the database, it will check/authenticate if the user has already registered and if the inputted username and password match the registered credentials.
4. Home Screen	After logging in, the Home Screen appears, offering buttons to access Single Player, Multiplayer, Leaderboards, and Logout. Clicking any button redirects the user to the corresponding screen.
5. Single Player	This section includes the Mechanics Screen, Character Selection Screen, and Count Screen. These appear sequentially after clicking the NEXT button.
⇒ Mechanics Screen	This page is for the game mechanics, wherein it contains 3 parts: Mechanics.kt, Mechanics2.kt, and Mechanics3.kt. These 3 pages will show up, respectively, after the button is clicked. In Mechanics.kt, this contains the corresponding points the fruit has. In Mechanics2.kt, this contains the overview of the snake's sprites or the character selection overview (not clickable yet). In Mechanics3.kt, this contains the additional reminders for the players to note before starting the game.
⇒ Character Selection Screen	This page contains 6 characters, which the player can already click/choose ONE character as the snake's head/sprite. If the user clicked the PROCEED button, it will not proceed to the next page (Count Screen) because it requires the player to choose a sprite. Upon clicking the character, a message will appear saying that 'that' character is selected.
⇒ Count Screen	After the user chooses a character, this page will appear. So, this page contains the countdown from 3 to 1 with a text saying Ready player and the <i>player's</i>

	<i>name</i> . After the countdown ends, it will proceed to the actual game already.
6. Game Screen	This contains the logo, game screen bg, and the snake's activity (moving and eating fruits). Also, the user can see his/her score live with his/her registered name. The snake eats the fruit with a corresponding point (represented as grades as mentioned in the mechanics). The first 5 fruits will be IP which is equivalent to +1 pt, 3.00 which is equivalent to +2 pt, 2.00 which is equivalent to +3 pt, 3.00 which is equivalent to +4 pt.
7. Game Over Screen	If the snake collides with itself or eats itself, the game ends, and this screen displays a "Game Over" message. The game ends after the snake consumes at least one fruit.
8. Multiplayer	Players can select this option from the Home Screen. It includes options to create a game, join an existing game, or return to the Home Screen.
⇒ Create Game Screen	The host device creates a game, generates a game ID, and allows other players to join.
⇒ Join Game	Players can join a game by entering the game ID generated by the host.
⇒ Back to Home Screen	If the player clicks this button, it will go back to the home screen if the player ever wants to play single-player, check the leaderboards, or log out.
9. Leaderboards	This screen displays the highest scores fetched from the database, motivating players to compete for the top position.
10. Logout	Users can log out, which will take them back to the Landing Screen. This allows a new player or unregistered user to start playing.

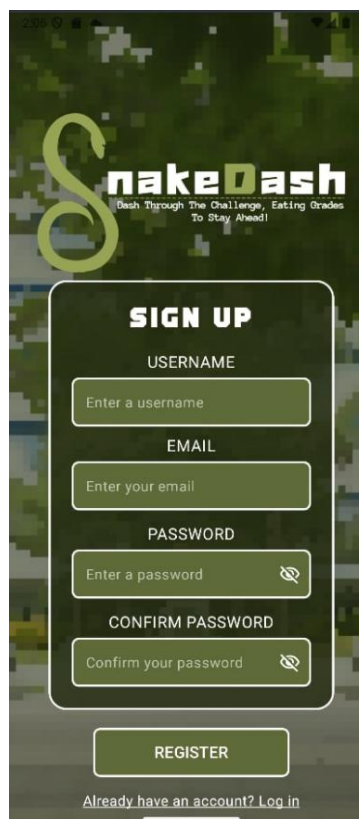
Table 4: Use Cases

UI/UX Designs

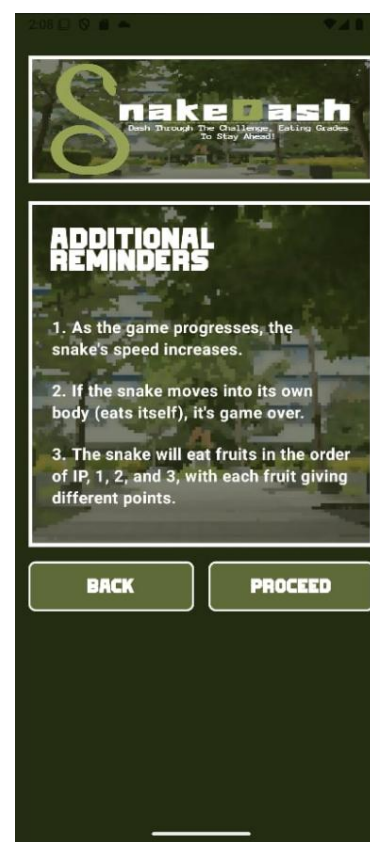
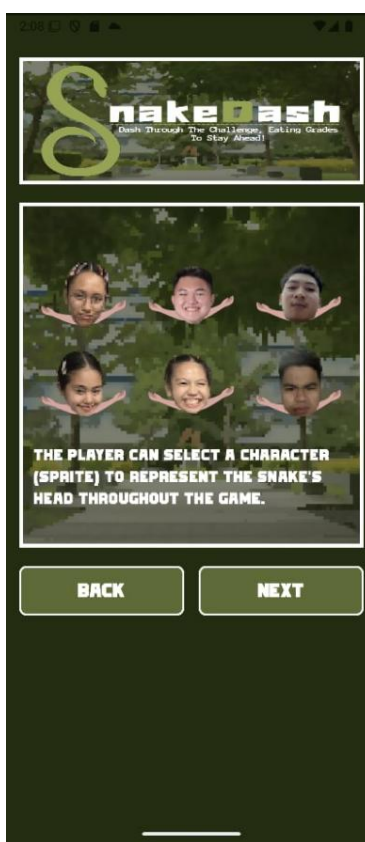
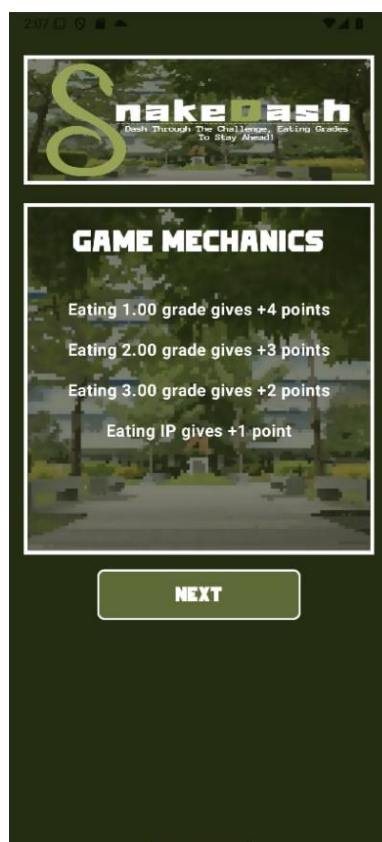
Landing Page



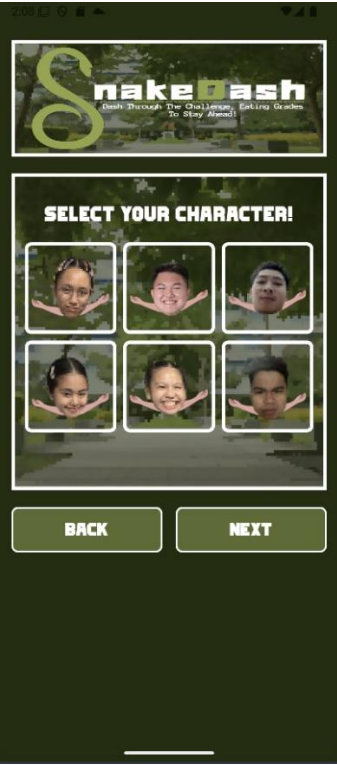
Login/SignUp Page



Mechanics Page



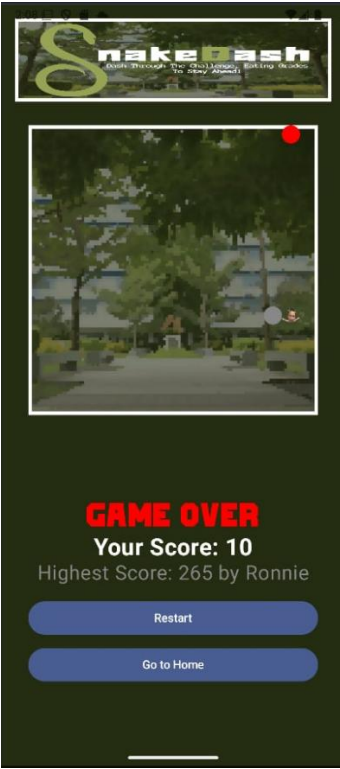
Character Selection Page



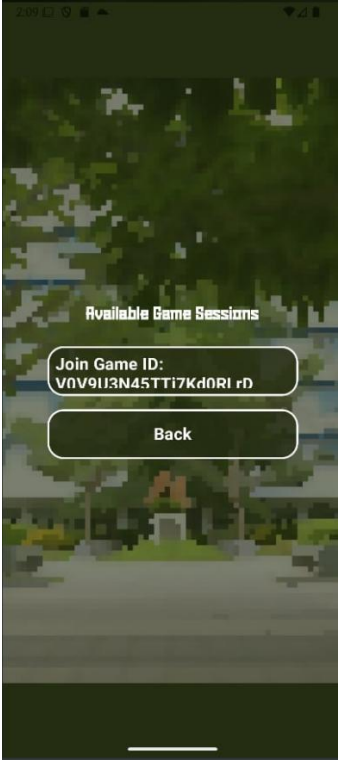
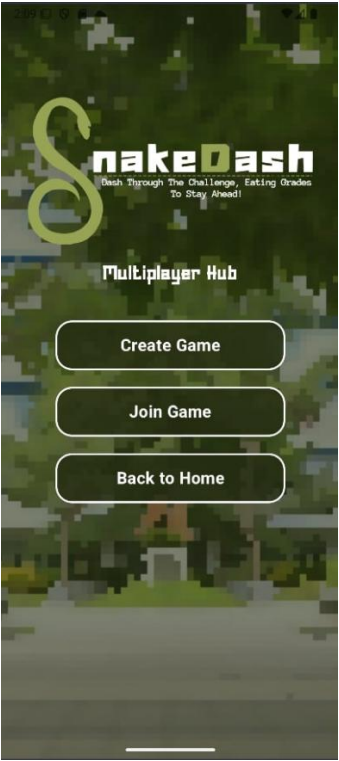
Game Screen



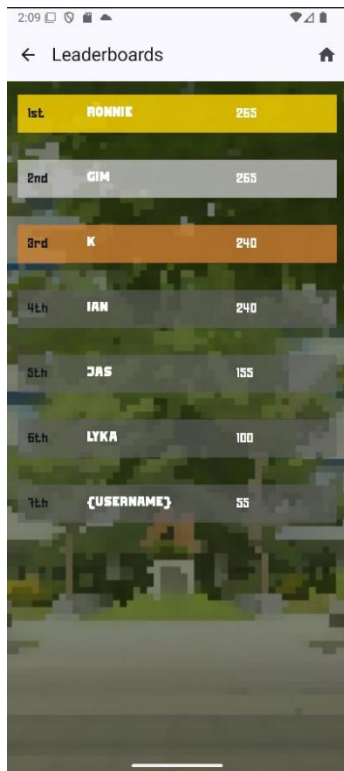
Game Over Page



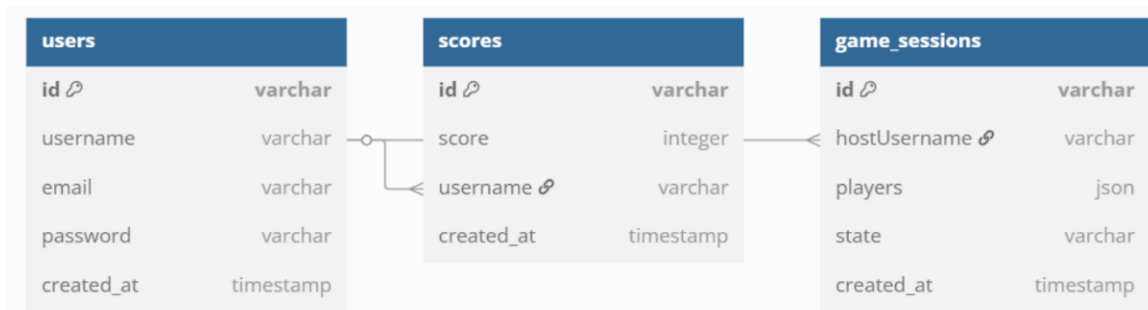
Multiplayer Page



Leaderboards Page



Database Design



Code Repository

<https://github.com/Edge-html/Final-Snake-Game-Repo.git>